Algebra 2 — Unit 2: Quadratic Functions

Practice Test — Study Copy — Answer Key

Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Vertex **C**
2. Axis of symmetry **F**
3. Zero (root) **E**
4. Discriminant **B**
5. Imaginary unit **D**
6. Completing the square **H**
7. Parabola **A**
8. Standard form **G**

(A) The U-shaped graph of a quadratic function.

(B) The expression $b^2 - 4ac$; its sign tells the number and type of solutions.

(C) The turning point of a parabola, where it reaches its maximum or minimum.

(D) The number $i = \sqrt{-1}$, which satisfies $i^2 = -1$.

(E) An x-value where $f(x) = 0$; an x-intercept of the parabola.

(F) The vertical line $x = h$ that splits a parabola into mirror halves.

(G) A quadratic written as $ax^2 + bx + c$.

(H) Rewriting $x^2 + bx$ as a perfect-square trinomial by adding $\left(\frac{b}{2}\right)^2$.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. The vertex of $y = (x-4)^2 - 1$ is **(A)** $(4, -1)$ ← (B) $(-4, -1)$ (C) $(4, 1)$ (D) $(-4, 1)$
2. $\sqrt{-25} =$ **(A)** $5i$ ← (B) -5 (C) $25i$ (D) 5
3. The discriminant of $x^2 - 4x + 7 = 0$ is -12, so the equation has (A) two real solutions (B) one real solution **(C) two complex (non-real) solutions** ← (D) no solutions of any kind
4. Factored completely, $x^2 - 49$ is **(A)** $(x-7)(x+7)$ ← (B) $(x-7)^2$ (C) $(x+7)^2$ (D) prime
5. The axis of symmetry of $y = x^2 - 6x + 5$ is **(A)** $x = 3$ ← (B) $x = -3$ (C) $x = 6$ (D) $x = 5$
6. A line and a parabola can intersect in *at most* how many points? (A) 1 **(B)** 2 ← (C) 3 (D) infinitely many

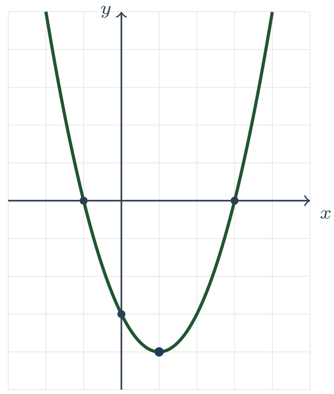
Teacher Note

Item 2: pull the i out first — $\sqrt{-25} = \sqrt{25}\,\sqrt{-1} = 5i$ (choice C forgets to simplify $\sqrt{25}$). Item 3: $b^2 - 4ac < 0 \Rightarrow$ the $\pm\sqrt{\ }$ is imaginary, so two complex conjugate roots and *zero* x-intercepts (D is wrong — there *are* solutions, just not real). Item 5: $x = -\frac{b}{2a} = \frac{6}{2} = 3$.

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The parabola f is shown below.



Vertex: $(1, -4)$; axis of symmetry: $x = 1$; y-intercept: $(0, -3)$;

zeros: $x = -1,\ x = 3$; minimum *or* maximum value: **minimum** -4; range: $y \geq -4$

2. **From standard form.** For $f(x) = x^2 - 6x + 5$, use $x = -\dfrac{b}{2a}$ to find the vertex, give the axis of symmetry, and state whether the parabola opens up or down. $x = -\dfrac{-6}{2(1)} = 3$**;** $f(3) = 9 - 18 + 5 = -4$**, so the vertex is** $(3, -4)$**. Axis of symmetry** $x = 3$**. Since** $a = 1 > 0$**, it opens *up* (so** -4 **is a minimum).**

3. **Solve by factoring.**

$$x^2 + 2x - 15 = 0$$

$(x + 5)(x - 3) = 0 \Rightarrow x = -5$ **or** $x = 3$**. Check:** $(-5)^2 + 2(-5) - 15 = 0$ ✓**,** $3^2 + 2(3) - 15 = 0$ ✓**.**

4. **Solve using square roots.**

$$(x - 2)^2 = 49$$

$x - 2 = \pm 7 \Rightarrow x = 2 + 7 = 9$ **or** $x = 2 - 7 = -5$**.**

5. **Solve by completing the square.**

$$x^2 + 6x - 7 = 0$$

$x^2 + 6x = 7$**; add** $\left(\frac{6}{2}\right)^2 = 9$ **to both sides:** $(x + 3)^2 = 16$**;** $x + 3 = \pm 4$**, so** $x = 1$ **or** $x = -7$**.**

6. **Complex numbers.** Let $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$. Write each answer in $a + bi$ form.

(a) $z_1 + z_2 = 4 - 2i$ (b) $z_1 - z_2 = 2 + 6i$

(c) $z_1 \cdot z_2 = (3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2 = 3 - 10i + 8 = 11 - 10i$

7. **Quadratic formula & discriminant.** Solve $2x^2 + 3x - 2 = 0$. First compute the discriminant and state how many real solutions to expect, then solve. $b^2 - 4ac = 3^2 - 4(2)(-2) = 9 + 16 = 25 > 0$**, so *two real* solutions.** $x = \dfrac{-3 \pm \sqrt{25}}{2(2)} = \dfrac{-3 \pm 5}{4}$**, giving** $x = \frac{1}{2}$ **or** $x = -2$**.**

8. **System.** Solve the linear–quadratic system algebraically. Give every solution as an ordered pair.

$$y = x^2 - 2x - 3 \qquad y = x - 3$$

Set equal: $x^2 - 2x - 3 = x - 3 \Rightarrow x^2 - 3x = 0 \Rightarrow x(x - 3) = 0$**, so** $x = 0$ **or** $x = 3$**. Then** $y = x - 3$**:** $x = 0 \Rightarrow (0, -3)$**;** $x = 3 \Rightarrow (3, 0)$**. Two solutions:** $(0, -3)$ **and** $(3, 0)$**.**

Teacher Note

Item 1: accept "opens up / minimum" language; range is $y \geq -4$ (closed, vertex included). Item 6c: full credit requires distributing all four terms *and* replacing $i^2 = -1$ — watch for students who leave $-8i^2$. Item 7: since the discriminant is a perfect square (25), the roots are rational; a non-perfect-square discriminant would leave a radical.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **Projectile.** A ball is thrown upward from a rooftop. Its height (in feet) after t seconds is

$$h(t) = -16t^2 + 32t + 48.$$

(a) From which feature of the model do you read the ball's *starting height*? Give that height. **(b)** Find the *maximum* height and the time it occurs. Which feature of the graph is this? **(c)** When does the ball hit the ground? Which feature answers this, and why do you reject the other solution? **(a) The starting height is the y-intercept, $h(0) = 48$ feet. (b) The maximum is the *vertex*. $t = -\dfrac{b}{2a} = -\dfrac{32}{2(-16)} = 1$ second; $h(1) = -16 + 32 + 48 = 64$ feet. So the ball reaches a maximum height of 64 ft at $t = 1$ s. (c) The ground is $h = 0$ — the *zeros*. $-16t^2 + 32t + 48 = 0 \Rightarrow t^2 - 2t - 3 = 0 \Rightarrow (t-3)(t+1) = 0$, so $t = 3$ or $t = -1$. We reject $t = -1$ because a negative time is meaningless here; the ball lands at $t = 3$ seconds.**

2. **Discriminant reasoning.** Consider the equation $x^2 - 4x + k = 0$, where k is a constant. **(a)** Write the discriminant in terms of k. **(b)** For what values of k does the equation have two real solutions? one? two complex solutions? **(c)** Explain how the discriminant is connected to the number of x-intercepts of the parabola $y = x^2 - 4x + k$. **(a) $b^2 - 4ac = (-4)^2 - 4(1)(k) = 16 - 4k$. (b) Two real: $16 - 4k > 0 \Rightarrow k < 4$. One (double) real: $16 - 4k = 0 \Rightarrow k = 4$. Two complex: $16 - 4k < 0 \Rightarrow k > 4$. (c) The real solutions are exactly the x-values where the parabola meets the x-axis. So a positive discriminant ($k < 4$) means two x-intercepts, a zero discriminant ($k = 4$) means one (the vertex sits on the x-axis), and a negative discriminant ($k > 4$) means no x-intercepts (the whole parabola stays above the axis).**

Teacher Note

Scoring Part D (6 pts each). D1: 2 pts (a) y-intercept named *and* 48 ft; 2 pts (b) vertex $t = 1$, height 64 ft; 2 pts (c) zeros, $t = 3$, with the $t = -1$ rejection reasoned from context. D2: 1 pt (a) $16 - 4k$; 3 pts (b) all three cases with correct boundary $k = 4$; 2 pts (c) ties discriminant sign to the *count of x-intercepts*. Do not give full (c) credit for restating (b) without the graph connection.